MILITARY SPECIFICATION

CARTRIDGE, 105MM, CLEARING LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the US Army Armament Research and Development Command, Department of the Army, and is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This document covers the loading, assembling and packing for one type of Clearing Cartridge for use with the 105MM Howitzer M137.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-P-46565	- Primer Percussion and M28B2, Loading, Assembling and Packing

DRAWINGS

ARMY

TPO-01-073-9920	- Packing and Marking for Container, For 105MM How. Ammunition (USAF Use Only)
TPO-01-073-9920	- Container, For 105MM How. Ammo (USAF Use Only)

FSC: 1315

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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9277353	- Packing and Marking for Box, Pallet Type, Wirebound for 105MM How. Ammo W/Fuze (USAF Use Only)
9278402	- Box, Pallet Type, Wirebound, Packing, Ammo For 105MM W/Fuze (USAF Use Only)
9329506	- Cartridge, 105MM, Clearing

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Material and parts shall be in accordance with applicable drawings and specifications.

3.2 Cartridges.-The cartridges shall comply with all requirements specified on drawing (Dwg.) 9329506 and with all requirements specified in applicable specifications.

3.3 Proving ground.-The cartridge shall comply with the following requirements when tested as specified.

3.3.1 Safety.-There shall be no premature burst or indication of metal parts separation in the gun bore.

3.3.2 Performance.-The clearing round shall expel a prepositioned projectile from a gun tube.

3.4 Workmanship

3.4.1 Cartridge.-All manufacturing processing and assembly operations of the cartridge shall be correctly performed in a thoroughly workmanlike manner. Parts shall be free of: burrs, cracks; also, dirt, chips, grease, rust and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspection.-The following type of inspection shall be conducted on this item:

a. Quality Conformance Inspection

4.3 Quality conformance inspection

4.3.1 Inspection lot formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of Propellant Charge Assemblies shall contain:

a. Propellant from not more than one lot from one manufacturer.

b. Primers from lots with the same interfix number from one manufacturer.

c. Cartridge cases from lots with the same interfix number from one manufacturer.

4.3.2 Examination.-See MIL-A-48078.

a. Sampling plans.-Unless otherwise specified in the Classification of Defects and Test tables, sampling plan and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63344 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.1	Primer, Prior to Assembly to Case (See Tables covering a detail of dwg. 9329506)		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	One or more flash holes missing		0.40%	3.2	Visual
Major	None defined				
Minor	None defined				

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63344 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.2	Cartridge Case, Prior to Assembling Propelling Charge (See dwg. 8595386 covering a detail of dwg. 9329506)		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Primer missing or damaged		0.40%	3.2	Visual
Minor					
201	Foreign matter in interior		0.65%	3.2	Visual
202	Evidence of poor workmanship (see 3.4)		0.65%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63344 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.3	Propelling Charge, Prior to Assembly to Cartridge Case (See dwg. 9288695 covering a detail of dwg. 9329506)		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any opening in bag large enough to permit escape of propellant		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.4	Cartridge Case With Propellant Charge Prior to Assembling Closing Cup (See Dwg. 9329506)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Foil side of bag not facing primer		0.40%	3.2	Visual
Minor	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63344 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.5	Cartridge Assembly (See Dwg. 9329506)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Primer above flush				Gage
Major					
101	Any component missing		0.40%	3.2	Visual
102	Depth to closing cup		0.40%	3.2	Manual
103	Retainer loose		0.40%	3.2	Manual
Minor					
201	Marking misleading, unidentifiable or missing		0.65%	3.2	Visual
202	Evidence of poor workmanship (see 3.4)		0.65%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63344 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.6	Sealed Container				TPO-01-073-9920
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Container damaged		0.40%	3.2	Visual-Manual
102	Metal end loose or distorted		0.40%	3.2	Visual-Manual
103	Material incomplete or badly wrinkled		0.40%	3.2	Visual
Minor					
201	Contents loose		0.65%	3.2	Gage
202	Gap between cover and body container		0.65%	3.2	Manual
203	Marking misleading, missing or unidentifiable		0.65%	3.2	Visual

NOTES:

6

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63344 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.7	Sealed Pallet Type Wirebound (See Dwg. 9278402 and 9277353) if applicable		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box damaged exposing contents		0.40%	3.2	Visual
102	Hardware or strapping missing, broken or loose		0.40%	3.2	Visual-Manual
103	DOD symbol missing or incorrect		0.40%	3.2	Visual
104	Board broken		0.40%	3.2	Visual
Minor					
201	Strapping or hardware improperly assembled		0.65%	3.2	Visual-Manual
202	Handle missing or insecure		0.65%	3.2	Visual-Manual
203	Car seal missing, unsealed or improperly positioned		0.65%	3.2	Visual
204	Marking misleading, missing or unidentifiable		0.65%	3.2	Visual

NOTES:

10

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4.3.3 Testing

4.3.3.1 Profile and alignment (see dwg. 9329506) - major defect.-The cartridge shall be gaged 100% for profile and alignment. Any cartridge which fails to comply with the applicable requirement shall be classed defective and removed from lot.

4.3.3.2 Proving ground

4.3.3.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements, fifteen sample cartridges shall be randomly selected from each lot for this test. If any clearing cartridge fails to comply with the requirements, the lot shall be rejected.

4.3.3.2.2 After three consecutive lots have met the criteria of 4.3.3.2.1, five sample cartridges shall be randomly selected from each lot for this test. If any clearing cartridge fails to comply with the requirements, the lot shall be rejected.

4.3.3.3 Check test for possible deterioration of Primer, Percussion, M28, (see specification MIL-P-46565). If the total elapsed time between original acceptance of the M28 primer lot and the assembly of that lot into complete round cartridges exceeds two years, or if the M28 primers have been subjected to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in MIL-P-46565 immediately before the M28 primer lot is assembled into the cartridges.

4.3.3.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.3.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6 herein.

4.4 Test methods and procedures

4.4.1 Proving ground.-These tests shall be conducted at a government proving ground in accordance with the applicable acceptance test procedure. The sample cartridges shall be fired in a M137 gun after being conditioned to 70 degrees Fahrenheit (^{o}F) $\pm$ 2.5^{o}F (21.11^{o} Centigrade (C) $\pm$ 1.39^{o}C). Prior to firing an inert loaded M1 projectile shall be seated into the forcing cone of the weapon.

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5. PACKAGING

5.1 Packaging requirements.-Packaging shall be as specified on Dwg. TPO-01-073-9920.

5.2 Packing and marking.-Packing and marking shall be as specified on Dwgs. TPO-01-073-9920.

6. NOTES

6.1 Intended use.-The components covered by this specification are intended for use in the M137 Howitzer Cannon.

6.2 Ordering data.-See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals.-See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, NJ 07801.

6.4 Cost of check test.-The contracting officer shall arrange for the contractor to be reimbursed for the expense incurred in the performance of the check tests. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the cartridge, and shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

6.5 Submission of test data.-In addition to the normal distribution of records, when the cartridge is procured by the Department of the Army, the address for forwarding ballistic test data is:

Commander
ARRADCOM
ATTN: DRDAR-QAR-E
Dover, New Jersey 07801

Custodian:
ARMY-AR

Preparing Activity:
Army-AR

Project No.: 1315-A229

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-63344 Cartridge. 105MM, Clearing LAP

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

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